

ANSIBLE_TASK_1

What is Ansible?

Ansible is an **automation tool** used to:

- Configure servers
- Install software
- Deploy applications
- Manage infrastructure

 It works using **YAML playbooks** and **agentless architecture**.

Importance of Ansible

1 Agentless (Very Important)

- No agent/software needed on worker nodes
- Uses **SSH (Linux)** and **WinRM (Windows)**
- Easy to manage thousands of servers

 Less maintenance

 More secure

2 Simple & Human-Readable

- Written in **YAML**
- Easy to learn compared to scripts

Example:

- name: Install nginx

apt:

name: nginx

state: present

Even non-programmers can understand it 

3 Automation Saves Time

Instead of manually:

- Logging into 100 servers
- Running same commands

Ansible does it in **one command**.

 Faster deployments

 No human errors

4 Idempotent Nature

Means:

Running the same playbook multiple times will not break anything.

If software already exists → Ansible skips it.

This is very important for **production stability**.

5 Used with DevOps Tools

Ansible integrates with:

- Jenkins
- Docker
- Kubernetes
- AWS / Azure / GCP

That's why Ansible is widely used in **CI/CD pipelines**.

Ansible Architecture

Ansible Controller

|

SSH / WinRM

|

| | |
Worker Worker Worker
Node Node Node

What is Ansible Controller?

The **controller** is the main machine where:

- Ansible is installed
- Playbooks are written
- Inventory is stored
- Commands are executed

Controller contains:

- Playbooks (YAML)
- Inventory file
- Roles
- Variables
- SSH keys

 Example:

Your laptop or Jenkins server can be controller.

What are Worker Nodes?

Worker nodes are:

- Target servers
- EC2 instances
- Linux / Windows machines

They do **not** need Ansible installed.

They only need:

- SSH access

- Python (usually preinstalled)
-

Importance of Controller–Worker Communication

This is **one of the most important concepts in Ansible** 🙌

1 Uses Secure Communication (SSH)

Controller → Worker communication happens via:

- **SSH (port 22)** for Linux
- **WinRM** for Windows

Benefits:

- Encrypted
 - Secure
 - No agents required
-

2 Push-Based Model

Ansible uses **push model**:

- Controller pushes tasks to workers
- Workers just execute commands

Unlike Puppet/Chef (pull model).

-  Better control
 -  Immediate execution
-

3 Inventory-Based Management

Controller uses inventory file:

[web]

server1

server2

[db]

server3

This allows:

- Group-wise execution
 - Environment separation (dev, qa, prod)
-

Scalability

One controller can manage:

- 10 servers
- 100 servers
- 10,000 servers

Because communication is lightweight SSH.

Centralized Control

Only controller has:

- Logic
- Credentials
- Playbooks

Workers never store sensitive data.

 Better security model.

Flow of Communication (Easy)

1. Controller reads playbook
2. Controller checks inventory
3. Controller connects to worker via SSH
4. Executes module temporarily
5. Gets result
6. Disconnects

No permanent service running on worker ❌

📌 Real-Time Example

You want to install nginx on 50 servers.

Without Ansible:

- Login 50 times
- Run same command

With Ansible:

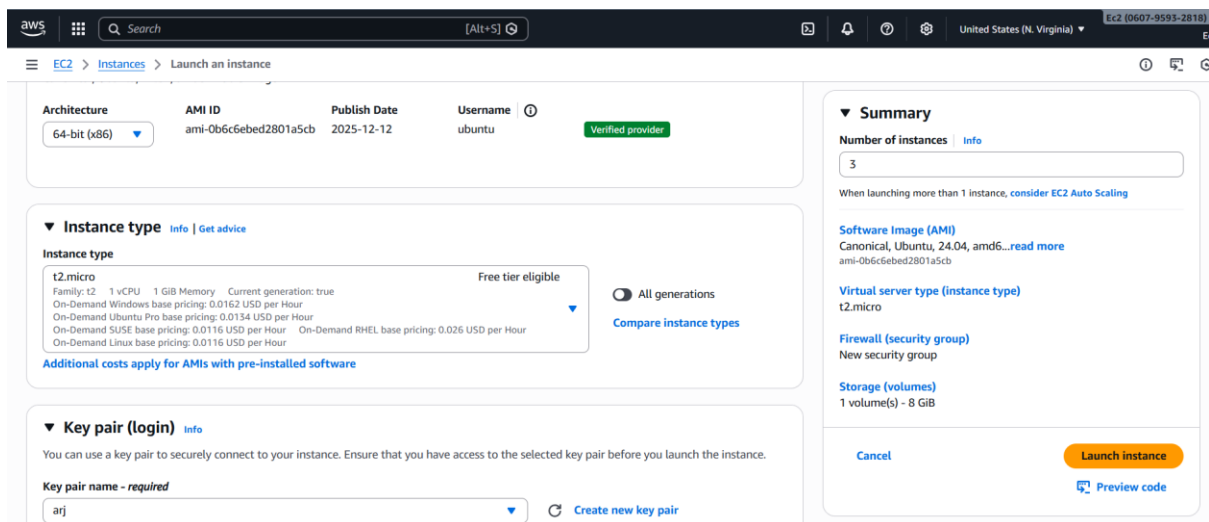
ansible-playbook install-nginx.yml

Done in seconds 🚀

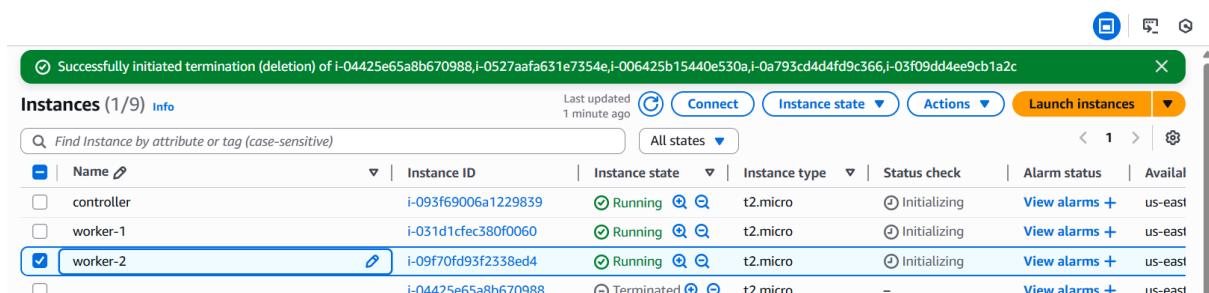
1. Watch ansible-01 video and write down notes.

2. Setup one master and two worker nodes in ansible.

→ Go to Aws console and Launch 3 instances



→ Three instances are now created, 1st one is given name as controller and second will be worker-1 and 2nd one will worker-2



	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availal
☐	controller	i-093f69006a1229839	Running	t2.micro	Initializing	View alarms +	us-east
☐	worker-1	i-031d1cfec380f0060	Running	t2.micro	Initializing	View alarms +	us-east
☒	worker-2	i-09f70fd93f2338ed4	Running	t2.micro	Initializing	View alarms +	us-east
☐		i-04425e65a8b670988	Terminated	t2.micro	-	View alarms +	us-east

→ After that update and install ansible

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-24-53:~$ sudo su -
root@ip-172-31-24-53:~# python3 --version
Python 3.12.3
```

→python3 --version

```
root@ip:172-31-24-53:~# python3 --version
python 3.12.3
root@ip:172-31-24-53:~# sudo apt update -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Fetched 16.4 MB in 1s (15.0 MB/s)
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
0 packages can be upgraded. New packages to be installed: 0.
To show suggested package upgrades, hold ALT (or F10) and press Y.
To show package details and suggestions, hold ALT (or F10) and press D.
```

→ apt install ansible

```
62 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-24-53:~# apt install ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfile python3-ntlm-auth python3-pas
  python3-simplejson python3-winnrm python3-xmlltodict
Suggested packages:
  --cowsay sshpass python3-trio python3-aiounic python3-h2 python3-httpx python3-httpcore python-lockfile-doc
```

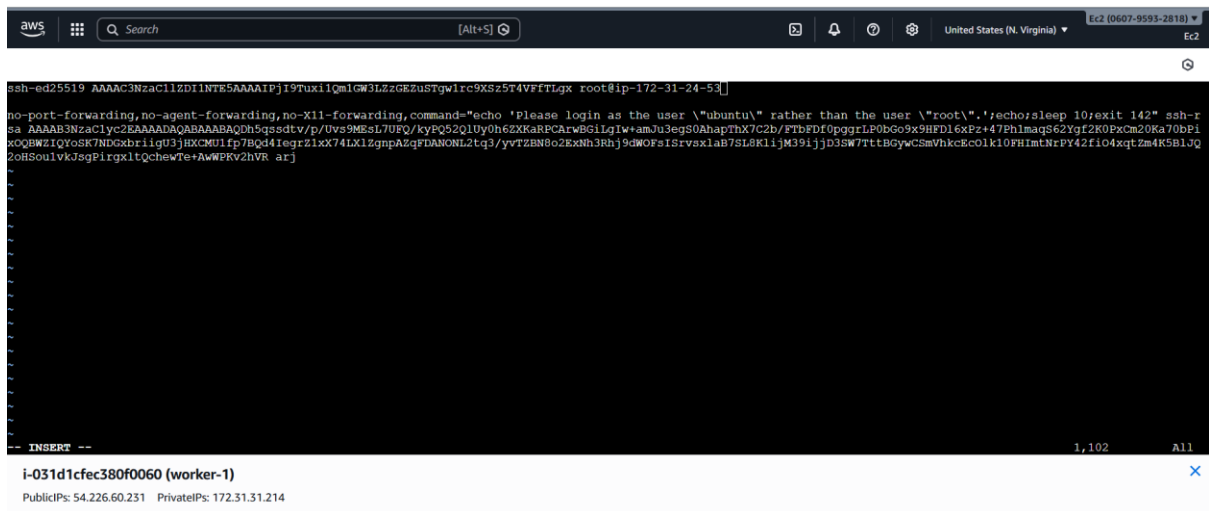
→ Go to host inventory and worker ips

```
root@ip-172-31-24-53: /etc/ansible
[all]
172.31.31.214
172.31.20.77
[worker-1]
172.31.31.214
[worker-2]
172.31.20.77
~
~
~
~
```

→Generate ssh key from controller

```
# root@ip-172-31-24-53:~# ssh-keygen  
Generating public/private ed25519 key pair.  
Enter file in which to save the key (/root/.ssh/id_ed25519):  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Your identification has been saved in /root/.ssh/id_ed25519  
Your public key has been saved in /root/.ssh/id_ed25519.pub  
The key fingerprint is:  
SHA256:HStFud8I3dWwMNPZ00uefXk+3zAQjSSmH/Qkw root@ip-172-31-24-53  
The key's randomart image is:  
  
--[ED25519 256]--+  
+.o.XX|  
..+...+|  
o.o.o.o=||  
..E.o.o.o||  
S....oo+||  
.....+||  
..o=||  
..oo||  
|[-----[SHA256]-----+  
root@ip-172-31-24-53:~# cat /root/.ssh/id_rsa_pub  
cat: /root/.ssh/id_rsa_pub: No such file or directory  
root@ip-172-31-24-53:~# cat /root/.ssh/id_rsa_pub  
cat: /root/.ssh/id_rsa_pub: No such file or directory  
root@ip-172-31-24-53:~# cat /root/.ssh/id_ed25519.pub  
ssh-ed25519 AAAACgPNlAHzatCZDILNtSAAAPAI91uxU1loMlw3LZZGeZusTgwlrKXSZ5t4VflGx root@ip-172-31-24-53  
root@ip-172-31-24-53:~#
```

→Add ssh key in worker-1 “vi .ssh/authorized_keys”



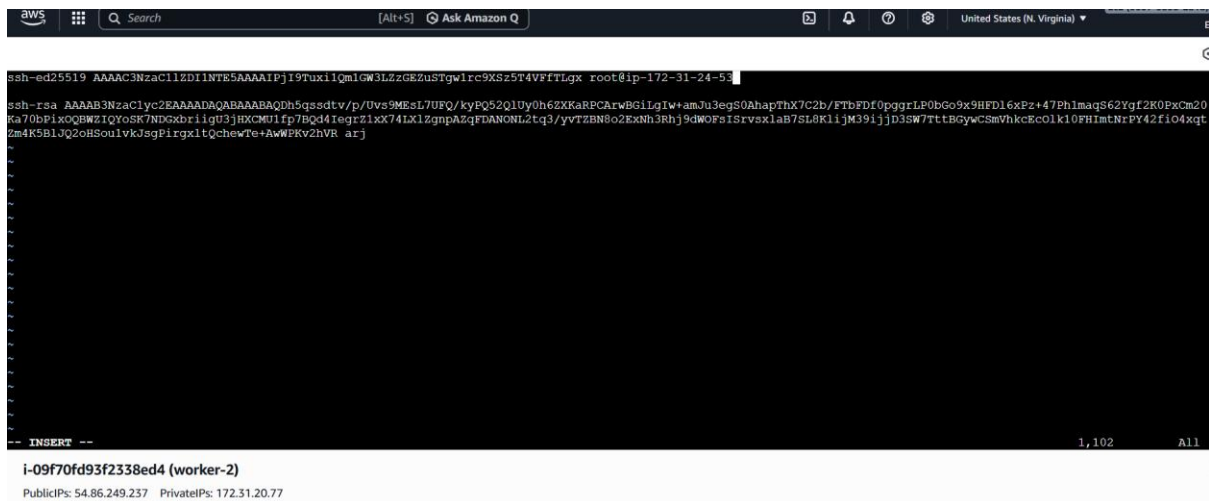
The screenshot shows the AWS Management Console interface. At the top, there's a search bar and navigation icons. Below, a terminal window for instance **i-031d1cfec380f0060 (worker-1)** is displayed. The terminal shows a successful SSH connection to a Linux instance. The user is **root** at **ip-172-31-24-53**. The terminal output includes a long SSH key being added to the **authorized_keys** file. The instance details below the terminal show Public IP: 54.226.60.231 and Private IP: 172.31.31.214.

→Give permission as 600



This screenshot shows the terminal for the same worker-1 instance. The user has navigated to the **.ssh** directory and used the **vi** editor to edit **authorized_keys**. The terminal shows the command **chmod 600 authorized_keys** being executed successfully. The instance details below show Public IP: 54.226.60.231 and Private IP: 172.31.31.214.

→Add ssh key in worker-2 “vi .ssh/authorized_keys”



The screenshot shows the AWS Management Console interface for instance **i-09f70fd93f2338ed4 (worker-2)**. The terminal window shows a successful SSH connection to a Linux instance. The user is **ubuntu** at **ip-172-31-20-77**. The terminal output includes a long SSH key being added to the **authorized_keys** file. The instance details below the terminal show Public IP: 54.86.249.237 and Private IP: 172.31.20.77.

→Give permission as 600



This screenshot shows the terminal for the worker-2 instance. The user has navigated to the **.ssh** directory and used the **vi** editor to edit **authorized_keys**. The terminal shows the command **chmod 600 authorized_keys** being executed successfully. The instance details below show Public IP: 54.86.249.237 and Private IP: 172.31.20.77.


```

root@ip-172-31-24-53:~# ssh root@172.31.31.214
The authenticity of host '172.31.31.214 (172.31.31.214)' can't be established.
ED25519 key fingerprint is SHA256:LEWlh0P/ia+x02Zqs5C93BY2nx5Wk6Jnddizpc0qfVE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.31.214' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jan 28 09:50:42 UTC 2026

System load:  0.0          Processes:      119
Usage of /:   26.4% of 6.71GB   Users logged in: 1
Memory usage: 22%          IPv4 address for enx0: 172.31.31.214
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@ip-172-31-31-214:~#

```

→ Check connection using command from controller “ansible all -m ping

```

root@ip-172-31-24-53:~# ansible all -m ping
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
172.31.31.214 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
172.31.20.77 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
root@ip-172-31-24-53:~#

```

3. Execute the adhoc command shared in #dvps-cloud-documents-12 Channel.

To check the connetivity between slaves

→ ansible all -m ping

→ansible all -m ping -i ansible_hosts

```

root@ip-172-31-24-53:~# cat /root/arj_inventory
[arj]
172.31.20.77
root@ip-172-31-24-53:~# cd
root@ip-172-31-24-53:~# ansible -i /root/arj_inventory arj -m ping
172.31.20.77 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
root@ip-172-31-24-53:~#

```

How to gather facts of slave machine

→ansible all -m setup

```

root@ip-172-31-24-53:~# ansible all -m setup
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
172.31.20.77 | success: {
  "ansible_facts": {
    "ansible_all_ipv4_addresses": [
      "172.31.20.77"
    ],
    "ansible_all_ipv6_addresses": [
      "fe80::8ff:efff:fe9c:a671"
    ],
    "ansible_apparmor": {
      "status": "enabled"
    },
    "ansible_architecture": "x86_64",
    "ansible_bios_date": "08/24/2006",
    "ansible_bios_vendor": "Xen",
    "ansible_bios_version": "4.11.amazon",
    "ansible_board_asset_tag": "NA",
    "ansible_board_name": "NA",
    "ansible_board_serial": "NA",
    "ansible_board_vendor": "NA",
    "ansible_board_version": "NA",
    "ansible_chassis_asset_tag": "NA",
    "ansible_chassis_serial": "NA",
    "ansible_chassis_vendor": "Xen",
    "ansible_chassis_version": "NA",
    "ansible_endline": {
      "BOOT_IMAGE": "/vmlinuz-6.14.0-1018-aws",
      "console": "ttyS0",
      "hwme_core_io_timeout": "4294967295",
      "panic": "-1",
      "ro": true,
      "root": "PARTUUID=2baf9194-a5fb-4b12-adb0-45166f0feb06"
    },
    "ansible_date_time": {
      "date": "2026-01-28",

```

→ansible all -a uptime

Tips: -m is the module and -a should contain the command it should run which goes as an argument to command and shell.

check the free memory or memory usage of host

→ansible all -a "free -m"

Execute a command as root user (sudo) on host

→ansible all -m shell -a "cat /etc/passwd|grep -i ansadm" -s --ask-sudo-pass

```

root@ip-172-31-24-53:~# ansible all -a uptime
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
172.31.31.214 | CHANGED | rc=0 >>
10:44:37 up 1:32, 1 user, load average: 0.00, 0.01, 0.00
172.31.20.77 | CHANGED | rc=0 >>
10:44:37 up 1:32, 1 user, load average: 0.02, 0.05, 0.01
root@ip-172-31-24-53:~# ansible all -a "free -m"
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
172.31.20.77 | CHANGED | rc=0 >>
total        used        free      shared  buff/cache   available
Mem:           957          345          332           0          436          612
Swap:              0              0              0
172.31.31.214 | CHANGED | rc=0 >>
total        used        free      shared  buff/cache   available
Mem:           957          337          365           0          411          619
Swap:              0              0              0
root@ip-172-31-24-53:~# ansible all -m shell -a "cat /etc/passwd|grep -i ansadm" -s --ask-sudo-pass
usage: ansible [-h] [--version] [-v] [-b] [--become-method BECOME_METHOD] [--become-user BECOME_USER] [-k] [--become-password-file BECOME_PASSWORD_FILE] [--inventory INVENTORY] [--list-hosts] [--subset]
               [-P POLL_INTERVAL] [--seconds SECONDS] [--o] [--t TREE] [--private-key PRIVATE_KEY_FILE] [--remote-user REMOTE_USER] [--connection CONNECTION] [--t TIMEOUT] [--ssh-common-args SSH_COMMON_ARGS]
               [--sftp-extra-args SFTP_EXTRA_ARGS] [--scp-extra-args SCP_EXTRA_ARGS] [--ssh-extra-args SSH_EXTRA_ARGS] [-k] [--connection-password-file CONNECTION_PASSWORD_FILE] [--c] [--d] [-e EXTRA_VARS]
               [--vault-id VAULT_IDS] [--] [-i] [--vault-password-file VAULT_PASSWORD_FILES] [--f FORKS] [--m MODULE_PATH] [--playbook-dir BASEDIR] [--task-timeout TASK_TIMEOUT] [--a MODULE_ARGS] [--m MODULE_NAME]
               pattern
ansible: error: unrecognized arguments: -s --ask-sudo-pass
usage: ansible [-h] [--version] [-v] [-b] [--become-method BECOME_METHOD] [--become-user BECOME_USER] [-k] [--become-password-file BECOME_PASSWORD_FILE] [--inventory INVENTORY] [--list-hosts] [--subset]
               [-P POLL_INTERVAL] [--seconds SECONDS] [--o] [--t TREE] [--private-key PRIVATE_KEY_FILE] [--remote-user REMOTE_USER] [--connection CONNECTION] [--t TIMEOUT] [--ssh-common-args SSH_COMMON_ARGS]
               [--sftp-extra-args SFTP_EXTRA_ARGS] [--scp-extra-args SCP_EXTRA_ARGS] [--ssh-extra-args SSH_EXTRA_ARGS] [-k] [--connection-password-file CONNECTION_PASSWORD_FILE] [--c] [--d] [-e EXTRA_VARS]
               [--vault-id VAULT_IDS] [--] [-i] [--vault-password-file VAULT_PASSWORD_FILES] [--f FORKS] [--m MODULE_PATH] [--playbook-dir BASEDIR] [--task-timeout TASK_TIMEOUT] [--a MODULE_ARGS] [--m MODULE_NAME]
               pattern
define and run a single task 'playbook' against a set of hosts
positional arguments:
  pattern              host pattern
options:
  --become-password-file BECOME_PASSWORD_FILE, --become-pass-file BECOME_PASSWORD_FILE

```

```

root@ip-172-31-24-53:~# ansible all -m shell -a "cat /etc/passwd|grep -i ansadm" -s --ask-sudo-pass
usage: ansible [-h] [--version] [-v] [-b] [--become-method BECOME_METHOD] [--become-user BECOME_USER] [-k] [--become-password-file BECOME_PASSWORD_FILE] [--inventory INVENTORY] [--list-hosts] [--subset]
               [-P POLL_INTERVAL] [--seconds SECONDS] [--o] [--t TREE] [--private-key PRIVATE_KEY_FILE] [--remote-user REMOTE_USER] [--connection CONNECTION] [--t TIMEOUT] [--ssh-common-args SSH_COMMON_ARGS]
               [--sftp-extra-args SFTP_EXTRA_ARGS] [--scp-extra-args SCP_EXTRA_ARGS] [--ssh-extra-args SSH_EXTRA_ARGS] [-k] [--connection-password-file CONNECTION_PASSWORD_FILE] [--c] [--d] [-e EXTRA_VARS]
               [--vault-id VAULT_IDS] [--] [-i] [--vault-password-file VAULT_PASSWORD_FILES] [--f FORKS] [--m MODULE_PATH] [--playbook-dir BASEDIR] [--task-timeout TASK_TIMEOUT] [--a MODULE_ARGS] [--m MODULE_NAME]
               pattern
ansible: error: unrecognized arguments: -s --ask-sudo-pass
usage: ansible [-h] [--version] [-v] [-b] [--become-method BECOME_METHOD] [--become-user BECOME_USER] [-k] [--become-password-file BECOME_PASSWORD_FILE] [--inventory INVENTORY] [--list-hosts] [--subset]
               [-P POLL_INTERVAL] [--seconds SECONDS] [--o] [--t TREE] [--private-key PRIVATE_KEY_FILE] [--remote-user REMOTE_USER] [--connection CONNECTION] [--t TIMEOUT] [--ssh-common-args SSH_COMMON_ARGS]
               [--sftp-extra-args SFTP_EXTRA_ARGS] [--scp-extra-args SCP_EXTRA_ARGS] [--ssh-extra-args SSH_EXTRA_ARGS] [-k] [--connection-password-file CONNECTION_PASSWORD_FILE] [--c] [--d] [-e EXTRA_VARS]
               [--vault-id VAULT_IDS] [--] [-i] [--vault-password-file VAULT_PASSWORD_FILES] [--f FORKS] [--m MODULE_PATH] [--playbook-dir BASEDIR] [--task-timeout TASK_TIMEOUT] [--a MODULE_ARGS] [--m MODULE_NAME]
               pattern
define and run a single task 'playbook' against a set of hosts
positional arguments:
  pattern              host pattern
options:
  --become-password-file BECOME_PASSWORD_FILE, --become-pass-file BECOME_PASSWORD_FILE
  --become-password-file BECOME_PASSWORD_FILE, --become-pass-file BECOME_PASSWORD_FILE
  --connection-password-file CONNECTION_PASSWORD_FILE, --conn-pass-file CONNECTION_PASSWORD_FILE
  --connection-password-file CONNECTION_PASSWORD_FILE
  --list-hosts          outputs a list of matching hosts; does not execute anything else
  --playbook-dir BASEDIR
                        Since this tool does not use playbooks, use this as a substitute playbook directory. This sets the relative path for many features including roles/ group_vars/ etc.
  --task-timeout TASK_TIMEOUT
                        set task timeout limit in seconds, must be positive integer.
  --vault-id VAULT_IDS
                        the vault identity to use. This argument may be specified multiple times.
  --vault-password-file VAULT_PASSWORD_FILES, --vault-pass-file VAULT_PASSWORD_FILES
                        vault password file
  --version             show program's version number, config file location, configured module search path, module location, executable location and exit
  -B SECONDS, --background SECONDS
                        run asynchronously, failing after x seconds (default=N/A)
  -C, --check           don't make any changes; instead, try to predict some of the changes that may occur
  -D, --diff            when changing (Gmail) files and templates, show the differences in those files; works great with --check
  -J, --ask-vault-password, --ask-vault-pass
                        ask for vault password
  -K, --ask-become-pass
                        ask for privilege escalation password
  -M MODULE_PATH, --module-path MODULE_PATH
                        prepend colon-separated path(s) to module library (default=[{ ANSIBLE_HOME ~ "/plugins/modules:/usr/share/ansible/plugins/modules" }]). This argument may be specified multiple times.
  -P POLL_INTERVAL, --poll POLL_INTERVAL
                        set the poll interval if using -B (default=15)
  -a MODULE_ARGS, --args MODULE_ARGS
                        The action's options in space separated k=v format: -a 'opt1=opt2=opt3' or a json string: -a '{"opt1": "val1", "opt2": "val2"}'
  -e EXTRA_VARS, --extra-vars EXTRA_VARS

```

```
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
172.31.20.77 | CHANGED | rc=0 >>
172.31.31.214 | CHANGED | rc=0 >>
root@ip-172-31-24-53:~#
```

```
root@ip-172-31-31-214:~# ls
arcj  snap
root@ip-172-31-31-214:~#
```

i-031d1cfec380f0060 (worker-1)

PublicIPs: 54.226.60.231 PrivateIPs: 172.31.31.214

```
root@ip-172-31-20-77:~# ls
arj  snap
root@ip-172-31-20-77:~#
```

i-09f70fd93f2338ed4 (worker-2)

PublicIPs: 54.86.249.237 PrivateIPs: 172.31.20.77

```
→ansible all -m yum -a "name=httpd state=latest"
```

[illegible]

```

Last login: Wed Jan 28 09:43:36 2026 from 18.206.107.29
ubuntu@ip-172-31-31-214:~$ sudo su -
root@ip-172-31-31-214:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-28 10:59:32 UTC; 4min 50s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 5130 (apache2)
      Tasks: 55 (limit: 1121)
     Memory: 5.9M (peak: 6.1M)
        CPU: 52ms
    CGroup: /system.slice/apache2.service
            └─5130 /usr/sbin/apache2 -k start
            └─5133 /usr/sbin/apache2 -k start
            └─5134 /usr/sbin/apache2 -k start

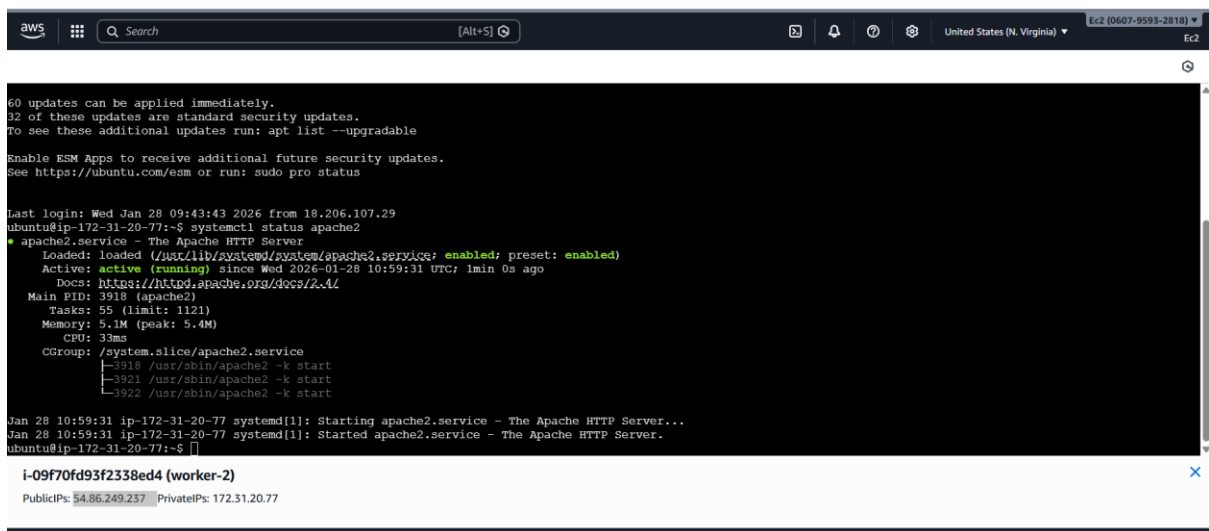
Jan 28 10:59:32 ip-172-31-31-214 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 10:59:32 ip-172-31-31-214 systemd[1]: Started apache2.service - The Apache HTTP Server.
root@ip-172-31-31-214:~# █

```

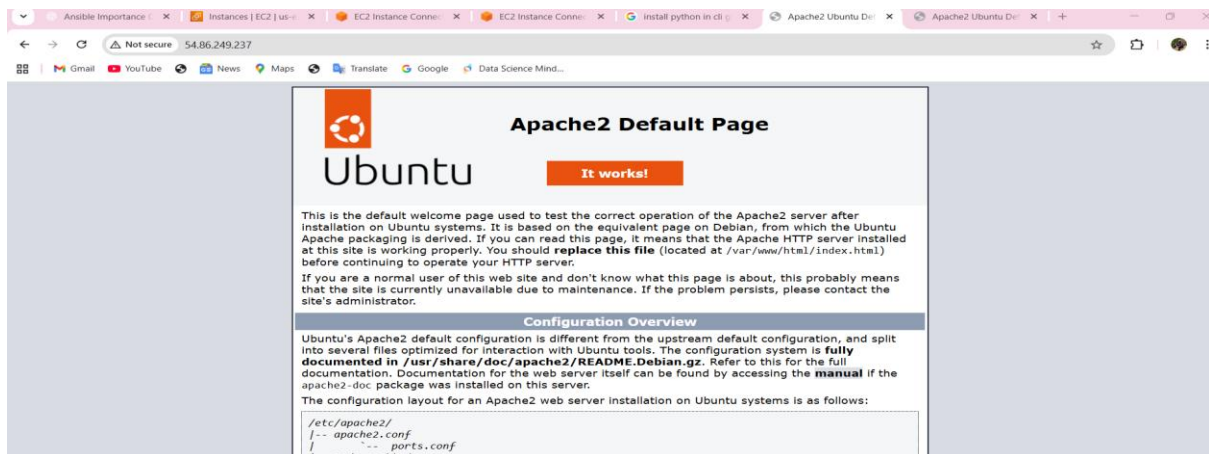
→confirm using public-Ip its is working



→when I checked the status using command “systemctl status apache2” its active



→confirm using public-Ip it is working



→ansible all -m service -a "name=apache2 state=stopped enabled=yes"

```
root@ip-172-31-24-53:~# ansible all -m service -a "name=apache2 state=stopped enabled=yes"
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
172.31.31.214 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "enabled": true,
  "name": "apache2",
  "state": "stopped",
  "status": {
    "ActiveEnterTimestamp": "wed 2026-01-28 10:59:32 UTC",
    "ActiveEnterTimestampMonotonic": "6459037381",
    "ActiveExitTimestampMonotonic": "0",
    "ActiveState": "active",
    "After": "basic.target network.target tmp.mount systemd-tmpfiles-setup.service system.slice nss-lookup.target -.mount remote-fs.target systemd-journald.socket sysinit.target",
    "AllowIsolate": "no",
    "AssertResult": "yes",
    "AssertTimestamp": "wed 2026-01-28 10:59:32 UTC",
    "AssertTimestampMonotonic": "6458965553",
    "Before": "apache-htcacheclean.service shutdown.target multi-user.target",
    "BlockIOAccounting": "no",
    "BlockIOWeight": "[not set]",
    "CPUAccounting": "yes",
    "CPUAffinityFromNUMA": "no",
    "CPUAffinityFromNUMA": "no",
    "CPUQuotaPerSec": "100%",
    "CPUQuotaPerSecUSec": "100000000",
    "CPUQuotaPeriodSec": "1s",
    "CPUSchedulingPolicy": "SCHED_OTHER",
    "CPUSchedulingPriority": "0",
    "CPUSchedulingResetInfork": "no",
    "CPUShares": "[not set]",
    "CPUUsageSec": "101395000",
    "CPUWeight": "[not set]",
    "CacheDirectoryMode": "0755",
    "CanFreeze": "yes"
  }
}
```

→Apache2 is inactive after stopping as we can see in worker-1

```
root@ip-172-31-31-214:~# systemctl status apache2
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: inactive (dead) since Wed 2026-01-28 11:13:07 UTC; 1min 53s ago
   Duration: 13min 35.193s
   Docs: https://httpd.apache.org/docs/2.4/
   Process: 6107 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 5130 (code=exited, status=0/SUCCESS)
   CPU: 101ms

Jan 28 10:59:32 ip-172-31-31-214 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 10:59:32 ip-172-31-31-214 systemd[1]: Started apache2.service - The Apache HTTP Server.
Jan 28 11:13:07 ip-172-31-31-214 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Jan 28 11:13:07 ip-172-31-31-214 systemd[1]: apache2.service: Deactivated successfully.
Jan 28 11:13:07 ip-172-31-31-214 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
```

i-031d1cfec380f0060 (worker-1)

PublicIPs: 54.226.60.231 PrivateIPs: 172.31.31.214

→Apache2 is inactive after stopping as we can see in worker-2

```
ubuntu@ip-172-31-20-77:~$ systemctl status apache2
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: inactive (dead) since Wed 2026-01-28 11:13:07 UTC; 1min 59s ago
   Duration: 13min 35.777s
   Docs: https://httpd.apache.org/docs/2.4/
   Process: 4070 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 3918 (code=exited, status=0/SUCCESS)
   CPU: 94ms

Jan 28 10:59:31 ip-172-31-20-77 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 10:59:31 ip-172-31-20-77 systemd[1]: Started apache2.service - The Apache HTTP Server.
Jan 28 11:13:07 ip-172-31-20-77 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Jan 28 11:13:07 ip-172-31-20-77 systemd[1]: apache2.service: Deactivated successfully.
Jan 28 11:13:07 ip-172-31-20-77 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
```

i-09f70fd93f2338ed4 (worker-2)

PublicIPs: 54.86.249.237 PrivateIPs: 172.31.20.77

To Start

→ansible all -m service -a "name=httpd state=started enabled=yes"

```
root@ip-172-31-24-53:~# ansible all -m service -a "name=apache2 state=started enabled=yes"
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
172.31.31.214 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "enabled": true,
  "name": "apache2",
  "state": "started",
  "status": {
    "ActiveEnterTimestamp": "wed 2026-01-28 10:59:32 UTC",
    "ActiveEnterTimestampMonotonic": "6459037381",
    "ActiveExitTimestamp": "wed 2026-01-28 11:13:07 UTC",
    "ActiveExitTimestampMonotonic": "7274231108",
    "ActiveState": "inactive",
    "After": "basic.target network.target tmp.mount systemd-tmpfiles-setup.service system.slice nss-lookup.target -.mount remote-fs.target systemd-journald.socket sysinit.target",
    "AllowIsolate": "no",
    "AssertResult": "yes",
    "AssertTimestamp": "wed 2026-01-28 10:59:32 UTC",
    "AssertTimestampMonotonic": "6458965553",
    "Before": "apache-htcacheclean.service shutdown.target multi-user.target",
    "BlockIOAccounting": "no",
    "BlockIOWeight": "[not set]",
    "CPUAccounting": "yes",
    "CPUAffinityFromNUMA": "no",
    "CPUAffinityFromNUMA": "no",
    "CPUQuotaPerSec": "100%",
    "CPUQuotaPerSecUSec": "100000000",
    "CPUQuotaPeriodSec": "1s",
    "CPUSchedulingPolicy": "SCHED_OTHER",
    "CPUSchedulingPriority": "0",
    "CPUSchedulingResetInfork": "no",
    "CPUShares": "[not set]",
    "CPUUsageSec": "101395000",
    "CPUWeight": "[not set]",
    "CacheDirectoryMode": "0755",
    "CanFreeze": "yes"
  }
}
```


→when I checked the status using command “systemctl status apache2” its active and started

```
Jan 28 10:59:32 ip-172-31-31-214 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 10:59:32 ip-172-31-31-214 systemd[1]: Started apache2.service - The Apache HTTP Server.
Jan 28 11:13:07 ip-172-31-31-214 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Jan 28 11:13:07 ip-172-31-31-214 systemd[1]: apache2.service: Deactivated successfully.
Jan 28 11:13:07 ip-172-31-31-214 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
root@ip-172-31-31-214:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-28 11:17:55 UTC; 1min 55s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Process: 6340 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 6343 (apache2)
      Tasks: 55 (limit: 1121)
     Memory: 5.3M (peak: 5.5M)
        CPU: 31ms
    CGroup: /system.slice/apache2.service
            └─6343 /usr/sbin/apache2 -k start
              └─6345 /usr/sbin/apache2 -k start
                └─6346 /usr/sbin/apache2 -k start

Jan 28 11:17:55 ip-172-31-31-214 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 11:17:55 ip-172-31-31-214 systemd[1]: Started apache2.service - The Apache HTTP Server.
root@ip-172-31-31-214:~#
```

i-031d1cfec380f0060 (worker-1)

PublicIPs: 54.226.60.231 PrivateIPs: 172.31.31.214

→when I checked the status using command “systemctl status apache2” its active and started

```
Jan 28 10:59:31 ip-172-31-20-77 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 10:59:31 ip-172-31-20-77 systemd[1]: Started apache2.service - The Apache HTTP Server.
Jan 28 11:13:07 ip-172-31-20-77 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Jan 28 11:13:07 ip-172-31-20-77 systemd[1]: apache2.service: Deactivated successfully.
Jan 28 11:13:07 ip-172-31-20-77 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-20-77:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-28 11:17:55 UTC; 3min 5s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Process: 5102 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 5105 (apache2)
      Tasks: 55 (limit: 1121)
     Memory: 5.1M (peak: 5.3M)
        CPU: 32ms
    CGroup: /system.slice/apache2.service
            └─5105 /usr/sbin/apache2 -k start
              └─5107 /usr/sbin/apache2 -k start
                └─5108 /usr/sbin/apache2 -k start

Jan 28 11:17:55 ip-172-31-20-77 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 11:17:55 ip-172-31-20-77 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-20-77:~$
```

i-09f70fd93f2338ed4 (worker-2)

PublicIPs: 54.86.249.237 PrivateIPs: 172.31.20.77